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GO 58 Pumping Lemma (For Regular Languages)

⇒ Pumping Lemma is used to prove that a Language is NOT REGULAR

⇒ It cannot be used to prove that a Language is Regular

Lemma:

If A is a Regular Language, then A has a Pumping Length 'P' such that any string 'S' where $|S| \geq P$ may be divided into 3 parts $S = xyz$, such that the following conditions must be true:

- (1) $xy^iz \in A$ for every $i \geq 0$
- (2) $|y| > 0$
- (3) $|xy| \leq P$

To prove that a language is not Regular using Pumping Lemma, follow the below steps:

- Assume A is Regular
- It has to have a Pumping Length (say P)
- All strings longer than P can be pumped
 $|S| \geq P$
- Now find a string ' S ' in A such that
 $|S| \geq P$
- Divide S into xyz
- Show that $xy^iz \notin A$ for some i
- Then consider all ways that S can be divided into xyz
- show that none of these can satisfy all the 3 pumping conditions at the same time
- S cannot be Pumped == CONTRADICTION

by contradiction



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(61) Pumping Lemma Example 1 5.5

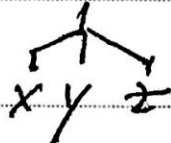
Using Pumping Lemma prove that the language $A = \{a^n b^n \mid n \geq 0\}$ is not Regular

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Proof:

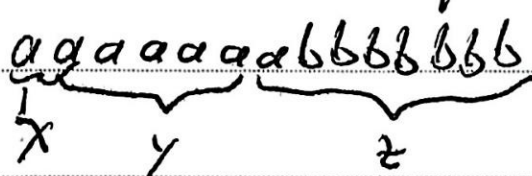
→ Assume that A is regular language

Pumping Length P , $S = a^P b^P$, $|S| = 2P$



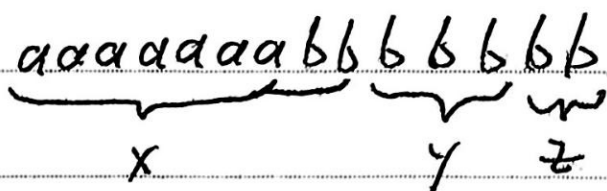
assume $P=7$, $S = aaaaaaa bbbbbb$

case 1: The Y is in 'a' part $xy^iz \Rightarrow xy^2z$



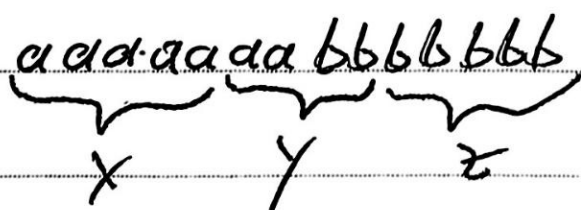
$aa aaaaaa aaaa abbb$
 $bbbb$
 $= a^{17} b^7 \notin A$

case 2: The Y is in 'b' part $xy^iz = xy^2z$



$aaaaaaa aab bbbbbb$
 $bb = a^7 b^{12} \notin A$

case 3: The Y is in the 'a' and 'b' part



$xy^iz \Rightarrow xy^2z$

$aaaaaa aabb aabb bbbbbb$

→ show that $xy^iz \notin A$ $\notin A$



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$$|xy| \leq p \quad p=7$$

~~case 1~~ case 1 $|xy| \leq 7$ ✓

case 2 $|xy| > 7$

case 3 $|xy| > 7$

$\Rightarrow$ S can't be pumped ✓

(Q2) Pumping lemma Example

$A = \{yy \mid y \in \{0,1\}^+\}$ is not Regular

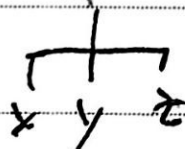
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Proof:

$\rightarrow$ Assume A is regular

Then it must have a pumping length P.

choose $S = 0^P 1 0^P 1$, $|S| \geq P$



~~assume $P=7$~~

xy^iz

⇒ 1. Assuming A is a regular language

2. there is $P > 0$ such that any string $w \in A$ with $|w| \geq P$ can be divided into three substring x, y , and z .

$$\begin{cases} w = xyz \end{cases}$$

$$|xy| \leq P$$

$$|y| > 0$$

for all $i \geq 0, xy^iz \in A$

3. choose $w = 0^P 1 0^P 1, |w| = 2P + 2 \geq P$

4. since $|xy| \leq P$, let's say $y = 0^k$ for some $k > 0$. Then $x = 0^j$ where $j + k \leq P$. And $z = 0^{P-j-k} 1 0^P 1$

5. let's choose $i = 1$. then $w = xz = 0^{P-k} 1 0^P 1$

obviously $w \notin A$. because $|y| > 0$, means $\boxed{k > 0}$
 $\neg \boxed{P-k < P}$

so w can't be pumped $\Rightarrow A$ is not a regular language.